

Mixed-Precision Randomized Sketching for Scaled Dot-Product Attention

Ján Kovačovský

Term Paper Draft ■ Mathematical Foundations of Machine Learning ■ 2026

Faculty of Mathematics and Physics, Charles University

Instructor: H. V. Le

Abstract. We propose a mathematically focused study of scaled dot-product attention under two simultaneous approximations: randomized feature sketching and mixed-precision arithmetic. The exact logit operator $d^{-1/2}QK^T$ is replaced by the sketched operator $d^{-1/2}QSS^TK^T$, where S is a random sketching matrix of smaller column dimension. The report isolates three levels of perturbation: the deterministic error in the logits, the induced perturbation of the rowwise softmax map, and the additional finite-precision error introduced when the sketched computation is carried out in low precision. The main analytical result is a clean error decomposition showing that the total output error is controlled by the sum of a sketching term and a rounding term. The accompanying experiments are designed as a validation of this decomposition rather than as a large-scale engineering benchmark.

Keywords: mixed-precision arithmetic, randomized numerical linear algebra, attention, transformers, sketching

1 Introduction

The purpose of this report is to study one analytically transparent component of the transformer architecture [1]: scaled dot-product attention. Two strands are particularly relevant. The first, rooted in randomized numerical linear algebra [2] [3], replaces the full attention matrix by a sketched surrogate [4] [5] [6]. The second, driven by hardware constraints, carries out the computation in reduced precision [7].

These two approximation strategies are typically studied in isolation. This report asks what happens when they are composed: when the query-key logit computation is approximated by a randomized sketch, can the subsequent computation be carried out in mixed precision with only a lower-order loss of accuracy?

The answer, as we show, is yes—and the mechanism is precise. Sketching reduces the effective dimension of the matrix product from d to s , which simultaneously controls the sketching error (via the intrinsic rank of the query-key subspace) and shrinks the finite-precision error (because the inner dimension of the product is smaller). The main result (Theorem 6) gives an explicit total error bound and identifies the condition $s \cdot u \ll \epsilon$ under which the finite-precision term is lower order, where s is the sketch dimension, u is the unit roundoff, and ϵ is the sketch tolerance. This condition provides

a concrete design rule: sketch dimension and precision should be chosen jointly, not independently.

2 Operator-Theoretic Setup

Let $\mathcal{H}_d := \mathbb{R}^d$ and $\mathcal{H}_n := \mathbb{R}^n$, both finite-dimensional Hilbert spaces endowed with their standard Euclidean inner products. We view matrices in $\mathbb{R}^{n \times d}$ as Hilbert–Schmidt operators

$$Q, K, V : \mathcal{H}_d \rightarrow \mathcal{H}_n.$$

Their Hilbert–Schmidt norm is denoted by $\|\cdot\|_{\text{HS}}$, while $\|\cdot\|_2$ denotes the spectral norm on linear operators.

The exact scaled logit matrix, viewed as an operator on $\mathcal{H}_n$, is

$$\mathcal{L}(Q, K) := d^{-1/2} Q K^* \in \mathcal{L}(\mathcal{H}_n),$$

and the rowwise softmax map is denoted by

$$\Sigma : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}.$$

Accordingly, the attention output is the nonlinear operator

$$\mathcal{A}(Q, K, V) := \Sigma(\mathcal{L}(Q, K))V.$$

Now let $S \in \mathbb{R}^{d \times s}$ with $s < d$. The sketched logit operator is

$$\tilde{\mathcal{L}}_S(Q, K) := d^{-1/2} Q S S^* K^*.$$

The operator $P_S := S S^*$ is positive semidefinite with rank at most s , but it is not in general an orthogonal projector unless the columns of S are orthonormal. The corresponding sketched attention output is

$$\tilde{\mathcal{A}}_S(Q, K, V) := \Sigma(\tilde{\mathcal{L}}_S(Q, K))V.$$

Thus the basic approximation step is the replacement of inner products $\langle q_i, k_j \rangle$ by sketched inner products $\langle S^* q_i, S^* k_j \rangle$.

3 Analytical Core

The analysis begins with a stability property of the rowwise softmax map.

Proposition 1 (Rowwise softmax is Hilbert–Schmidt Lipschitz). Let $L, M \in \mathbb{R}^{n \times n}$. Then

$$\|\Sigma(L) - \Sigma(M)\|_{\text{HS}} \leq \|L - M\|_{\text{HS}}.$$

Consequently, for every $V \in \mathbb{R}^{n \times d}$,

$$\|\Sigma(L)V - \Sigma(M)V\|_{\text{HS}} \leq \|L - M\|_{\text{HS}} \|V\|_2.$$

Proof. Let $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the softmax map on vectors with Jacobian $D\sigma(x) = \text{Diag}(p) - pp^T$ where $p = \sigma(x)$. For any $z \in \mathbb{R}^n$, $z^T D\sigma(x) z = \sum_i p_i z_i^2 - (\sum_i p_i z_i)^2 \leq \|z\|_2^2$, so $\|D\sigma(x)\|_2 \leq 1$ for all x . The mean value theorem gives $\|\sigma(x) - \sigma(y)\|_2 \leq \|x - y\|_2$; applying this row by row and squaring yields the HS bound. The second assertion follows from $\|XV\|_{\text{HS}} \leq \|X\|_{\text{HS}} \|V\|_2$. $\square$

The next proposition isolates the deterministic sketching error. Since attention logits are bilinear forms $q_i^* k_j$, the relevant notion is that S approximately preserves inner products on the row space of Q and K .

Proposition 2 (Deterministic sketching bound). Let $U \subseteq \mathcal{H}_d$ be the subspace spanned by the rows of $[Q^T \ K^T]$, and let $r := \dim U$. Assume that S is an ϵ -subspace embedding for U , meaning

$$|\langle S^*u, S^*v \rangle - \langle u, v \rangle| \leq \epsilon \|u\|_2 \|v\|_2 \quad \text{for all } u, v \in U.$$

Equivalently, if $U_0 \in \mathbb{R}^{d \times r}$ is any orthonormal basis matrix for U , then $\|U_0^T S S^* U_0 - I_r\|_2 \leq \epsilon$. Then

$$\|\mathcal{L}(Q, K) - \tilde{\mathcal{L}}_S(Q, K)\|_{\text{HS}} \leq \frac{\epsilon}{\sqrt{d}} \|Q\|_{\text{HS}} \|K\|_2.$$

Consequently,

$$\|\mathcal{A}(Q, K, V) - \tilde{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{\epsilon}{\sqrt{d}} \|Q\|_{\text{HS}} \|K\|_2 \|V\|_2.$$

Proof. Let $U_0 \in \mathbb{R}^{d \times r}$ have orthonormal columns spanning the joint row space of Q and K . Write $Q = Q_U U_0^T$ and $K = K_U U_0^T$, where $Q_U = Q U_0 \in \mathbb{R}^{n \times r}$ and $K_U = K U_0 \in \mathbb{R}^{n \times r}$. Then

$$QK^* - QSS^*K^* = Q_U(I_r - U_0^T S S^* U_0)K_U^T.$$

The subspace-embedding condition is exactly $\|U_0^T S S^* U_0 - I_r\|_2 \leq \epsilon$, so

$$\|QK^* - QSS^*K^*\|_{\text{HS}} \leq \|Q_U\|_{\text{HS}} \|I_r - U_0^T S S^* U_0\|_2 \|K_U^T\|_2 \leq \epsilon \|Q\|_{\text{HS}} \|K\|_2,$$

where we used $\|Q_U\|_{\text{HS}} = \|Q\|_{\text{HS}}$ and $\|K_U\|_2 = \|K\|_2$. Multiplying by $d^{-1/2}$ gives the logit bound. The attention-output bound follows from Proposition 1. $\square$

The proposition should be read as a conditional stability statement: it separates the attention-stability argument from the probabilistic question of when the sketch approximately preserves inner products on the relevant subspace.

The next result gives this probabilistic guarantee for Gaussian sketches. A scaled Gaussian matrix $S \in \mathbb{R}^{d \times s}$ with entries $\mathcal{N}(0, 1/s)$ is a subspace embedding for any fixed r -dimensional subspace with high probability, requiring $s = O(r/\epsilon^2)$ —not $O(d/\epsilon^2)$.

Proposition 3 (Probabilistic subspace embedding guarantee). Let $U \subseteq \mathcal{H}_d$ be an r -dimensional subspace. Let $S \in \mathbb{R}^{d \times s}$ have i.i.d. entries $S_{ij} \sim \mathcal{N}(0, 1/s)$. For

$\epsilon \in (0, 1)$ and $\delta \in (0, 1)$, if

$$s \geq C \frac{r + \ln(1/\delta)}{\epsilon^2},$$

for a universal constant C , then with probability at least $1 - \delta$, S is an ϵ -subspace embedding for U . If U contains the rows of $[Q^T \ K^T]$, then with the same probability,

$$\|\mathcal{A}(Q, K, V) - \tilde{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{\epsilon}{\sqrt{d}} \|Q\|_{\text{HS}} \|K\|_2 \|V\|_2.$$

Proof. Let $U_0 \in \mathbb{R}^{d \times r}$ be a matrix whose columns form an orthonormal basis for U . Then S is an ϵ -subspace embedding for U if and only if $\|U_0^T S S^* U_0 - I_r\|_2 \leq \epsilon$. The matrix $U_0^T S S^* U_0 = (S^* U_0)^* (S^* U_0)$ has the same distribution as $G^* G / s$ where $G \in \mathbb{R}^{s \times r}$ has i.i.d. standard normal entries. Standard JL concentration [2] gives $\Pr(\|G^* G / s - I_r\|_2 > \epsilon) \leq 2 \exp(-c s \epsilon^2 + r)$ for a universal constant $c > 0$, provided s is large enough relative to r/ϵ^2 . The stated condition with an appropriate C ensures the failure probability is at most δ . The attention error bound follows from Proposition 2. $\square$

The key feature is the dependence on r rather than d : when the query-key rows lie near a low-dimensional subspace, a much smaller sketch suffices.

We now turn to the finite-precision error. The standard Higham-type forward-error analysis [8] for the matrix product AB with $A \in \mathbb{R}^{n \times s}$, $B \in \mathbb{R}^{s \times n}$ gives

$$|\widehat{AB} - AB| \leq \gamma_s |A| |B|, \quad \gamma_s := \frac{su}{1 - su},$$

where u is the unit roundoff. For the sketched logit product $d^{-1/2}(QS)(KS)^*$, the inner dimension is s , so the finite-precision error is proportional to $s \cdot u$ —not $d \cdot u$ as it would be for the full product QK^* .

Proposition 4 (Finite-precision error for sketched logits). Let $\hat{\mathcal{L}}_S(Q, K)$ denote the logit matrix computed in floating-point arithmetic with unit roundoff u , and let $\gamma_s := su/(1 - su)$ for $su < 1$. Then

$$\|\tilde{\mathcal{L}}_S(Q, K) - \hat{\mathcal{L}}_S(Q, K)\|_{\text{HS}} \leq \frac{\gamma_s \|QS\|_{\text{HS}} \|KS\|_{\text{HS}}}{\sqrt{d}}.$$

Consequently,

$$\|\tilde{\mathcal{A}}_S(Q, K, V) - \hat{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{\gamma_s \|QS\|_{\text{HS}} \|KS\|_{\text{HS}} \|V\|_2}{\sqrt{d}}.$$

Proof. Write the sketched logit product as $d^{-1/2}AB^*$ with $A = QS \in \mathbb{R}^{n \times s}$ and $B = KS \in \mathbb{R}^{s \times n}$. The standard forward-error bound for matrix multiplication [8] gives

$$|\widehat{AB}^* - AB^*| \leq \gamma_s |A| |B^*|,$$

entrywise in absolute value. Taking the Frobenius (Hilbert–Schmidt) norm and applying the Cauchy–Schwarz inequality for matrices,

$$\|\widehat{AB}^* - AB^*\|_{\text{HS}} \leq \gamma_s \| |A| |B^*| \|_{\text{HS}} \leq \gamma_s \|A\|_{\text{HS}} \|B\|_{\text{HS}}.$$

$\curvearrowright$

↳ Multiplying by $d^{-1/2}$ yields the logit bound. The attention-output bound follows from Proposition 1. $\square$

Remark 5 (Scope of the finite-precision bound). The bound in Proposition 4 models the error in the final sketched logit product $(QS)(KS)^*$, assuming that QS and KS are given exactly. The experiments measure a broader implementation error that also includes quantization of Q , K , V , and S to the storage dtype before projection. Thus the measured E_{fp} is a broader implementation error than the multiplication-only term analyzed in the proposition. $\lrcorner$

The factor γ_s scales linearly with s : sketching reduces the finite-precision error by a factor of s/d compared to the unsketched product. This is the formal statement of the intuition that “compressing first makes low precision safer.” Combining the probabilistic subspace embedding guarantee with the finite-precision error analysis yields the main result.

Theorem 6 (Mixed-precision sketched attention). Under the hypotheses of Propositions 3 and 4, with probability at least $1 - \delta$,

$$\|\mathcal{A}(Q, K, V) - \hat{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{1}{\sqrt{d}} \left(\epsilon \|Q\|_{\text{HS}} \|K\|_2 + \gamma_s \|QS\|_{\text{HS}} \|KS\|_{\text{HS}} \right) \|V\|_2.$$

The first term is the sketching error; the second is the finite-precision error. The condition $s \cdot u \ll \epsilon$ is the leading-order requirement for the finite-precision term to remain lower order, assuming the norm factors are of comparable scale.

Proof. By Proposition 1 and the triangle inequality,

$$\|\mathcal{A} - \hat{\mathcal{A}}_S\|_{\text{HS}} \leq (\|\mathcal{L} - \tilde{\mathcal{L}}_S\|_{\text{HS}} + \|\tilde{\mathcal{L}}_S - \hat{\mathcal{L}}_S\|_{\text{HS}}) \|V\|_2.$$

Apply Proposition 2 to the first term and Proposition 4 to the second. For the lower-order claim, observe that $\gamma_s = O(s \cdot u)$ and $\|QS\|_{\text{HS}}, \|KS\|_{\text{HS}} = O(\sqrt{\|Q\|_2 \|K\|_2})$, so the finite-precision term is $O(s \cdot u \cdot \|Q\|_2 \|K\|_2 / \sqrt{d})$ while the sketching term is $\Theta(\epsilon \|Q\|_{\text{HS}} \|K\|_2 / \sqrt{d})$. When the norm factors are of comparable scale, the condition $s \cdot u \ll \epsilon$ ensures the finite-precision term is dominated. $\square$

Remark 7 (Design rule). Theorem 6 yields a concrete joint design rule. For float16 ($u_{16} \approx 5 \times 10^{-4}$) and typical sketch dimensions $s \leq 64$, we have $s \cdot u_{16} \leq 0.032$, which is smaller than typical sketch tolerances $\epsilon \geq 0.1$. For float32 ($u_{32} \approx 6 \times 10^{-8}$), the condition is essentially always satisfied. For float8 E4M3 ($u_{\text{fp8}} \approx 0.0625$), even $s = 16$ gives $s \cdot u_{\text{fp8}} \approx 1.0$, which is comparable to ϵ —so the finite-precision term is no longer clearly lower order. The rule also reveals a regime to avoid: if s is chosen too large relative to $1/u$, the finite-precision error can overtake the sketching error, defeating the purpose of mixed precision. $\lrcorner$

4 Numerical Experiments

The experiments probe the decomposition of Theorem 6 directly. For each combination of parameters (n, d, s) and each precision configuration we compute three logit matrices:

- (1) $\mathcal{L}(Q, K)$: exact unsketched logits in `float64` (reference);

- (2) $\tilde{\mathcal{L}}_S(Q, K)$: exact sketched logits in `float64`;
- (3) $\hat{\mathcal{L}}_S(Q, K)$: sketched logits in the target precision.

The sketching error is $E_{\text{sk}} := \|\mathcal{L} - \tilde{\mathcal{L}}_S\|_{\text{HS}}$ and the finite-precision error is $E_{\text{fp}} := \|\tilde{\mathcal{L}}_S - \hat{\mathcal{L}}_S\|_{\text{HS}}$. The total error is $E_{\text{tot}} := \|\mathcal{L} - \hat{\mathcal{L}}_S\|_{\text{HS}}$. The same decomposition is recorded for the attention output. We test four precision configurations: `fp32` (all stages in float32), `full_fp16` (all stages in float16), `mixed_fp16` (storage in float16, accumulation and softmax in float32), and `fp8_store` (storage in float8 E4M3, accumulation and softmax in float32).

The sketch $S \in \mathbb{R}^{d \times s}$ is a scaled Gaussian matrix with i.i.d. $\mathcal{N}(0, 1/s)$ entries; the scaling is chosen so that $\mathbb{E}[SS^*] = I_d$. We sweep $n \in \{64, 128, 256\}$, $d \in \{32, 64\}$, $s \in \{8, 16, 24, 32\}$, and average over three seeds. Data and sketch seeds are independent. The full numerical results are available in the committed CSV file `results/attention_sweep.csv`.

Table 8 reports absolute HS errors of the logits for $n = 256$, $d = 64$, together with the ratio $E_{\text{sk}}/E_{\text{fp}}$; the output errors showed the same qualitative ordering and are omitted.

Config	s	E_{sk}	E_{fp}	E_{tot}	$E_{\text{sk}}/E_{\text{fp}}$
fp32	8	7.22e2	1.2e-4	7.22e2	5.9e6
	16	5.03e2	1.1e-4	5.03e2	4.4e6
	24	4.27e2	9.4e-5	4.27e2	4.5e6
	32	3.70e2	9.1e-5	3.70e2	4.1e6
full_fp16	8	7.22e2	4.1e-1	7.22e2	1.8e3
	16	5.03e2	2.9e-1	5.03e2	1.8e3
	24	4.27e2	2.6e-1	4.27e2	1.7e3
	32	3.70e2	2.3e-1	3.70e2	1.6e3
mixed_fp16	8	7.22e2	3.8e-1	7.22e2	1.9e3
	16	5.03e2	2.6e-1	5.03e2	1.9e3
	24	4.27e2	2.4e-1	4.27e2	1.8e3
	32	3.70e2	2.1e-1	3.70e2	1.8e3
fp8_store	8	7.22e2	5.0e1	7.21e2	1.4e1
	16	5.03e2	3.4e1	5.01e2	1.5e1
	24	4.27e2	3.0e1	4.26e2	1.4e1
	32	3.70e2	2.6e1	3.70e2	1.4e1

Table 8 / Hilbert–Schmidt errors of the logit matrix ($n = 256$, $d = 64$, averaged over 3 seeds).
 $E_{\text{sk}} = \|\mathcal{L} - \tilde{\mathcal{L}}_S\|_{\text{HS}}$; $E_{\text{fp}} = \|\tilde{\mathcal{L}}_S - \hat{\mathcal{L}}_S\|_{\text{HS}}$; $E_{\text{tot}} = \|\mathcal{L} - \hat{\mathcal{L}}_S\|_{\text{HS}}$; $\|\mathcal{L}\|_{\text{HS}} \approx 256$.

Three observations stand out. First, for `fp32` and `fp16`, $E_{\text{sk}}/E_{\text{fp}}$ is consistently large (10^3 – 10^6), confirming that the dominant error source is the randomized sketch. The relative sketching error exceeds 140%, while the relative finite-precision error is below 0.16%. Second, `full_fp16` and `mixed_fp16` have finite-precision errors of the same order, consistent with Proposition 4: storage quantization (the $s \cdot u$ factor) dominates any benefit from higher-precision accumulation. Third, `fp8` storage substantially increases the finite-precision error relative to `fp16` (from ~ 0.3 to ~ 30 in absolute terms, or from $\sim 0.1\%$ to $\sim 10\%$ relative), but the sketching error still dominates by a factor of ~ 14 – 15 . That is, `fp8` storage degrades accuracy noticeably compared to `fp16`, yet in the

tested synthetic regime the total error remains sketching-dominated. A regime where finite precision dominates would require even lower precision or larger s ; the current experiments do not reach it. Thus the results support the error decomposition and illustrate the degradation predicted by Theorem 6 for increasing $s \cdot u$, but they do not demonstrate a full inversion of the error ratio.

5 Outlook

The present report studies only the feature-sketch model $QK^* \approx QSS^*K^*$. The natural extensions are Nyström-type approximations and random-feature approximations of the softmax kernel, which would connect the project more directly with the efficient-attention literature [5] [6]. Another direction is to replace the Gaussian sketch by a structured random matrix (e.g., a subsampled randomized Hadamard transform), which would yield the same subspace-embedding guarantees with lower computational cost. A possible extension would replace finite-dimensional matrices by Hilbert–Schmidt operators and study convergence under compactness or spectral-decay assumptions; we do not pursue this here.

References

Back-references to the pages where the publication was cited are given by .

- [1] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention Is All You Need. In: *Advances in Neural Information Processing Systems 30*. **2017**.
ARXIV: [1706.03762 \[cs.CL\]](#) 1
- [2] Nathan Halko, Per-Gunnar Martinsson, and Joel A. Tropp. Finding Structure with Randomness: Probabilistic Algorithms for Constructing Approximate Matrix Decompositions. *SIAM Review*, **2011**.
DOI: [10.1137/090771806](#) 1, 4
- [3] Michael W. Mahoney. Lecture Notes on Randomized Linear Algebra. **2016**.
ARXIV: [1608.04481 \[cs.DS\]](#) 1
- [4] Sinong Wang, Belinda Z. Li, Madian Khabsa, Han Fang, and Hao Ma. Linformer: Self-Attention with Linear Complexity. **2020**.
ARXIV: [2006.04768 \[cs.LG\]](#) 1
- [5] Yunyang Xiong, Zhanpeng Zeng, Rudrasis Chakraborty, Mingxing Tan, Glenn Fung, Yin Li, and Vikas Singh. Nyströmformer: A Nyström-Based Algorithm for Approximating Self-Attention. **2021**.
ARXIV: [2102.03902 \[cs.CL\]](#) 1, 7
- [6] Krzysztof Choromanski, Valerii Likhoshesterov, David Dohan, Xingyou Song, Andreea Gane, Tamas Sarlos, Peter Hawkins, Jared Davis, Afroz Mohiuddin, Lukasz Kaiser, David Belanger, Lucy Colwell, and Adrian Weller. Rethinking Attention with Performers. **2021**.
ARXIV: [2009.14794 \[cs.LG\]](#) 1, 7
- [7] Paulius Micikevicius, Sharan Narang, Jonah Alben, Gregory Diamos, Erich Elsen, David Garcia, Boris Ginsburg, Michael Houston, Oleksii Kuchaiev, Ganesh Venkatesh, and Hao Wu. Mixed Precision Training. In: *International Conference on Learning Representations*. **2018**. 1
- [8] Nicholas J. Higham. Accuracy and Stability of Numerical Algorithms. 2nd ed. SIAM, **2002**.
DOI: [10.1137/1.9780898718027](#) 4